CSA0919 PROGRAMMING IN JAVA
HARD LEVEL PROGRAMS (16-30)

ROSENIVITHA J

192424009

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```java
Main.java
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int m = sc.nextInt();
        int n = sc.nextInt();

        int[][] mat = new int[m][n];

        for(int i=0;i<m;i++){
            for(int j=0;j<n;j++){
                mat[i][j] = sc.nextInt();
            }
        }

        int k = sc.nextInt();

        int[][] arr = new int[m][2];

        for(int i=0;i<m;i++){
            int count=0;
            for(int j=0;j<n;j++){
                if(mat[i][j]==1) count++;
            }
            arr[i][0]=count;
            arr[i][1]=i;
        }

        Arrays.sort(arr,(a,b)->{
            if(a[0]==b[0]) return a[1]-b[1];
            return a[0]-b[0];
        });

        for(int i=0;i<k;i++){
            System.out.print(arr[i][1]+" ");
        }
    }
}
```

input

```
2 2
1 0
1 1
1
0
```

17

```java
import java.util.*;

public class Main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int num = sc.nextInt();

        int steps = 0;

        while(num > 0){
            if(num % 2 == 0)
                num /= 2;
            else
                num -= 1;

            steps++;
        }

        System.out.println(steps);
    }
}
```

input

```
5
4
```

18

```java
        System.out.println("Total= " + (int)total);
        System.out.printf("Aggregate = %.2f\n", avg);

        if(avg>75)
            System.out.println("Class: DISTINCTION");
        else if(avg>=60)
            System.out.println("Class: FIRST DIVISION");
        else if(avg>=50)
            System.out.println("Class: SECOND DIVISION");
        else if(avg>=40)
            System.out.println("Class: THIRD DIVISION");
        else
            System.out.println("Class: FAIL");
    }
}

public class Main {
    public static void main(String[] args){
        Student s = new Student();
        s.input();
        s.calculate();
    }
}
```

input

```
80 80 80 80 80 80
Total= 480
Aggregate = 80.00
Class: DISTINCTION
```

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```java
public class Main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int income = sc.nextInt();
        int tax = 0;

        if(income <= 250000){
            tax = 0;
        }
        else if(income <= 500000){
            tax = (int)((income - 250000) * 0.10);
        }
        else if(income <= 1000000){
            tax = (int)(250000*0.10 + (income - 500000)*0.20);
        }
        else{
            tax = (int)(250000*0.10 + 500000*0.20 + (income - 1000000)*0.30);
        }

        System.out.println("Tax= " + tax);
    }
}
```

input

```
300000
Tax= 5000
```

20

```java
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int consumerNo = sc.nextInt();
        String name = sc.next();
        int prev = sc.nextInt();
        int curr = sc.nextInt();

        int units = curr - prev;
        double bill = 0;

        if(units <= 100)
            bill = units * 1;
        else if(units <= 200)
            bill = 100*1 + (units-100)*2.5;
        else if(units <= 500)
            bill = 100*1 + 100*2.5 + (units-200)*4;
        else
            bill = 100*1 + 100*2.5 + 300*4 + (units-500)*6;

        System.out.println("Bill Amount: " + bill);
    }
}
```

input

```
1
Ram
100
150
Bill Amount: 50.0
```

21

```java
}

class Truck extends Vehicle {
    int load;

    void truckDetails(){
        Scanner sc = new Scanner(System.in);
        load = sc.nextInt();
    }

    void show(){
        display();
        System.out.println("Load: " + load);
    }
}

public class Main {
    public static void main(String[] args){
        Car c = new Car();
        c.input();
        c.carDetails();
        c.show();
    }
}
```

input

```
Ravi TN01
5
Owner: Ravi
License: TN01
Seats: 5
```

22

```java
    int id;

    void input(){
        Scanner sc = new Scanner(System.in);
        name = sc.next();
        address = sc.next();
        phone = sc.next();
        mobile = sc.next();
        head = sc.next();
        id = sc.nextInt();
    }

    void display(){
        System.out.println(name + " " + address + " " + phone + " " + mobile + " " + head + " " + id);
    }
}

public class Main {
    public static void main(String[] args){
        Person p = new Person();
        p.input();
        p.display();
    }
}
```

input

```
Ram Chennai 0441234 9876543210 Father 101
Ram Chennai 0441234 9876543210 Father 101
```

23

```java
        public void run(){
            System.out.println("Add: " + (10+5));
        }
    }

class Sub extends Thread {
    public void run(){
        System.out.println("Sub: " + (10-5));
    }
}

class Mul extends Thread {
    public void run(){
        System.out.println("Mul: " + (10*5));
    }
}

public class Main {
    public static void main(String[] args){
        new Add().start();
        new Sub().start();
        new Mul().start();
    }
}
```

input

```
Add: 15
Sub: 5
Mul: 50
```

24

```java
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.next();

        int n = s.length();
        int[] dp = new int[n];

        for(int i=0;i<n;i++){
            if(isPal(s.substring(0,i+1))){
                dp[i] = 0;
            } else {
                dp[i] = i;
                for(int j=0;j<i;j++){
                    if(isPal(s.substring(j+1,i+1))){
                        dp[i] = Math.min(dp[i], dp[j] + 1);
                    }
                }
            }
        }

        System.out.println(dp[n-1]);
    }
}
```

input

```
aabbx
2
```

25

Main.java

```java
        int k = ones / 3;
        int i1=-1,i2=-1,i3=-1;
        int count=0;

        for(int i=0;i<n;i++){
            if(arr[i]==1){
                count++;
                if(count==1) i1=i;
                if(count==k+1) i2=i;
                if(count==2*k+1) i3=i;
            }
        }

        while(i3 < n && arr[i1]==arr[i2] && arr[i2]==arr[i3]){
            i1++; i2++; i3++;
        }

        if(i3 == n){
            System.out.println((i1-1) + " " + i2);
        } else {
            System.out.println("-1 -1");
        }
    }
}
```

input

```
5
1 0 1 0 1
0 3
```

26

```java
        }
        return res;
    }
}

public class Main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        String[] words = new String[n];

        for(int i=0;i<n;i++)
            words[i] = sc.next();

        String pref = sc.next();
        String suff = sc.next();

        WordFilter wf = new WordFilter(words);

        System.out.println(wf.f(pref, suff));
    }
}
```

input

```
4
apple apply ape april
ap le
0
```

27

```java
    int sumRange(int l, int r){
        int sum = 0;
        for(int i=l;i<=r;i++)
            sum += arr[i];
        return sum;
    }
}

public class Main {
    public static void main(String[] args){
        int[] nums = {1,3,5};

        NumArray obj = new NumArray(nums);

        System.out.println(obj.sumRange(0,2)); // 9

        obj.update(1,2);

        System.out.println(obj.sumRange(0,2)); // 8
    }
}
```

input

```
9
8
```

28

```java
        for(int i=0;i<n;i++)
            arr[i] = sc.nextInt();

        HashSet<Integer> result = new HashSet<>();
        HashSet<Integer> prev = new HashSet<>();

        for(int x : arr){
            HashSet<Integer> curr = new HashSet<>();
            curr.add(x);

            for(int y : prev){
                curr.add(y | x);
            }

            result.addAll(curr);
            prev = curr;
        }

        System.out.println(result.size());
    }
}
```

input

```
3
1 1 2
3
```

29

```java
        int i = a.length-1, j = b.length-1, carry = 0;

        while(i>=0 || j>=0 || carry!=0){
            int sum = carry;

            if(i>=0) sum += a[i--];
            if(j>=0) sum += b[j--];

            res.add(Math.abs(sum % 2));
            carry = -(sum / 2);
        }

        while(res.size()>1 && res.get(res.size()-1)==0)
            res.remove(res.size()-1);

        Collections.reverse(res);

        System.out.println(res);
    }
}
```

input

```
[1, 0, 0, 0, 0]
```

30

```java

            if(backtrack(s, list, i+1)) return true;

            list.remove(list.size()-1);
        }

        return false;
    }

    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        String s = sc.next();

        List<Integer> res = new ArrayList<>();

        if(backtrack(s, res, 0))
            System.out.println(res);
        else
            System.out.println("[]");
    }
}
```

input

```
1101111\
[]
```